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Biofuel production utilizing black soldier fly (*Hermetia illucens*): a sustainable approach for organic waste management

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A R T I C L E I N F O

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A B S T R A C T

This study investigates the potential of biodiesel production from Black Soldier Fly (BSF) larvae using organic waste. The aim of this research is to assess the feasibility of biodiesel production and its economic viability from BSF larvae. Specific objectives include analyzing organic waste quantity, moisture content, dry matter, BOD content, and biodiesel conversion efficiency. Our findings reveal that under various scales, biodiesel production from BSF larvae shows promise. For instance, in a small-scale urban biofuel plant, we estimate the production of approximately 122.73 liters of biodiesel per day from 1 ton of organic waste. Moreover, our calculations demonstrate the potential for medium-scale rural and large-scale industrial biofuel plants to produce 1,678.85 and 22,272.73 liters of biodiesel per day, respectively. These results highlight the significant biodiesel potential of BSF larvae. In conclusion, this study provides valuable insights into the biodiesel production potential of BSF larvae. It underscores the feasibility of utilizing organic waste for biodiesel production, contributing to sustainable biofuel development. Our research contributes to the knowledge base by offering quantitative assessments of biodiesel potential. We have incorporated quantitative data to provide a clearer understanding of the biodiesel potential and its economic implications.

1. Introduction

1.1. Problem statement

The increasing global demand for energy, coupled with the growing environmental concerns associated with conventional fossil fuels, has accelerated the search for sustainable and renewable energy sources. Biofuel, produced from organic matter or wastes, has emerged as one of the promising alternatives to traditional fossil fuels due to its renewability and potential for reduced greenhouse gas emissions. However, sustainable feedstock production for biofuel remains a challenge, necessitating innovative solutions to leverage the vast, underutilized organic waste resources [1-4].

The persistent growth in global energy consumption and the inherent environmental implications associated with traditional fossil fuels have triggered an international urgency to explore renewable energy sources. Biofuels, derived from organic materials or wastes, have emerged as a significant front-runner in this arena. The potential of biofuels to act as renewable substitutes for conventional fossil fuels while mitigating greenhouse gas emissions has been well-documented. Despite these promising prospects, the production of sustainable and cost-effective biofuel remains an intricate challenge. This is predominantly due to the requirement of substantial feedstock, which often competes with food crops for arable land and fresh water resources. This necessitates innovative and sustainable strategies to harness the potential of organic waste as an alternative and viable feedstock source [5-7].

1.2. Current ongoing solution

Current efforts in biodiesel production largely rely on traditional vegetable oil and animal fat feedstocks. While these sources have contributed to the biodiesel industry, they face limitations in terms of scalability, resource availability, and competition with food production. Consequently, there is a pressing need for innovative approaches to biodiesel production that address these limitations and offer sustainable alternatives [8-12].

1.3. Proposed solution in this work

In response to the challenges faced by conventional biodiesel

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production methods, this study explores the potential of Black Soldier Fly (BSF) larvae as a novel and sustainable source for biodiesel production. BSF larvae have garnered attention for their ability to efficiently convert organic waste materials into valuable resources, including biodiesel. This work aims to assess the feasibility of utilizing BSF larvae as a biodiesel feedstock and investigates the economic viability of various production scales. In this context, the Black Soldier Fly (*Hermetia illucens*, BSF), a non-pest insect species renowned for its exceptional bioconversion capacity, presents a unique opportunity. BSF larvae are capable of thriving on a wide array of organic waste substrates, ranging from food waste and agricultural byproducts to animal manure. This property makes them an effective tool in waste reduction, potentially lightening the load on traditional waste management systems. More interestingly, these larvae exhibit a high lipid content, a precursor for biodiesel production via the transesterification process [7,13,14].

Black Soldier Fly (*Hermetia illucens*, BSF), an insect species known for its impressive waste bioconversion capability, offers a compelling solution to this problem. BSF larvae thrive on a broad range of organic waste, including food waste, agricultural residues, and animal manure, effectively reducing the waste volume and potentially alleviating the burden on conventional waste management systems. Importantly, these larvae contain a high content of lipids, which can be transformed into biodiesel, a type of biofuel, through the transesterification process [15,16].

1.4. Summarized current research novelty and objective of this work

The novelty of this research lies in its innovative approach to biodiesel production through the utilization of BSF larvae, which represents a sustainable and eco-friendly alternative to traditional feedstocks. This study seeks to quantify the biodiesel production potential of BSF larvae under different scenarios, considering factors such as organic waste quantity, moisture content, dry matter, Biochemical Oxygen Demand (BOD) content, and biodiesel conversion efficiency. By doing so, it aims to provide valuable insights into the feasibility and economic viability of biodiesel production using BSF larvae as a feedstock.

Through this comprehensive analysis, it is aspired to not only demonstrate the practicality of using BSF larvae in biofuel production but also to emphasize its broader impacts on the sustainable energy landscape and the circular economy. This research endeavors to provide a novel perspective on waste management and energy production, potentially paving the way for a more sustainable and resource-efficient future [17,18].

This research aims to exploit the dual potential of BSF larvae for sustainable waste management and biofuel production. It will elucidate the process of rearing BSF, processing larvae into biodiesel, and analyze the economic and environmental implications of this approach [19,20].

It is established a detailed process from the rearing of BSF to the conversion of harvested larvae into biodiesel, providing a comprehensive understanding of each stage. Additionally, the economic feasibility of this approach is investigated by considering all associated costs, from initial setup and maintenance to the biodiesel production process. The environmental implications are also examined with a focus on the overall carbon footprint in comparison to traditional waste management methods [21,22].

2. Materials and methods

2.1. Site selection and facility setup

The first step towards establishing a biofuel plant is the selection of a suitable site. The site must have sufficient space for the cultivation of Black Soldier Flies (BSFs), waste storage, biofuel production, and storage facilities. The selected site must also comply with local and national environmental and zoning regulations [23-25].

2.2. Black soldier fly cultivation

A controlled environment is necessary for the successful breeding and cultivation of BSFs. The breeding area must be designed in a way that prevents escape and predation, and promotes optimal growth. Mature BSFs are introduced into this breeding area and are allowed to lay eggs on provided substrates. The substrate for BSF larvae typically comprises a mixture of organic wastes, which can be sourced from local waste management facilities or agricultural operations [26-28].

2.3. Harvesting of BSF larvae

BSF larvae are harvested once they have reached the pre-pupal stage, as this is when their fat content is highest. Harvesting can be facilitated through the use of ramps or other structures that the larvae naturally climb when they are ready to pupate [29-31].

2.4. Larvae processing

The harvested larvae are processed for biofuel production. The first step involves the extraction of lipids from the larvae. This can be done using various methods, but solvent extraction is the most commonly used. The extracted lipids then undergo a transesterification process to convert them into biodiesel. This process involves reacting the lipids with an alcohol (usually methanol) in the presence of a catalyst (usually a strong base like sodium hydroxide or potassium hydroxide) [32-34].

2.5. Biodiesel purification and storage

The biodiesel produced from the transesterification process is then purified to remove any remaining catalyst, glycerol, and other impurities. The biodiesel is then stored in appropriate storage tanks until it is ready for distribution [35-37].

2.6. Waste management

The remaining solid and liquid wastes from the larvae rearing and biodiesel production processes are also managed. Solid residues can be composted and used as a soil amendment in agricultural operations, while liquid wastes can be treated and reused within the facility or safely disposed of according to local regulations [38-40].

2.7. Economic and environmental impact analysis

An integral part of this project is the analysis of its economic and environmental impacts. Costs associated with the setup and operation of the plant, including equipment, labor, maintenance, and waste management, are calculated and compared with the revenues from the sale of biodiesel. The environmental impact, focusing on waste reduction and greenhouse gas emissions, is also evaluated to assess the sustainability of the project.

This research is grounded in established methodologies for biofuel production from Black Soldier Fly (BSF) larvae, as demonstrated in several key academic studies. These papers provide critical insights into various aspects of biofuel production, including bioconversion processes, lipid extraction for biodiesel, and economic feasibility.

2.7.1. Bioconversion of dairy manure by black soldier fly for biodiesel and sugar production

This study offers valuable data on the use of BSF larvae in converting dairy manure into biodiesel, which informs our approach to bioconversion processes [41].

2.7.2. From organic waste to biodiesel: black soldier fly, hermetia illucens, makes it feasible

The findings from this research on the high fat content of BSF larvae are crucial for our methodologies in lipid extraction and biodiesel production [42] .

2.7.3. Enhancing the biodiesel yield: rearing black soldier fly larvae on solid residual fraction of restaurant waste

This paper's exploration of using restaurant waste for BSF larvae rearing aids in understanding the versatility of feedstock options for larvae growth [43].

2.7.4. Biodiesel production from rice straw and restaurant waste employing black soldier fly assisted by microbes

The co-conversion process discussed in this study helps in optimizing our approach for biodiesel production from varied organic waste [44].

2.7.5. Bioconversion of rice straw waste by black soldier fly larvae

Insights from this study guide our rate of feed and biomass production, contributing to the efficiency of our bioconversion process [45].

These referenced studies are pivotal in shaping the methodologies applied in this research, offering a comprehensive and scientifically grounded approach to utilizing BSF larvae for sustainable biofuel production.

The three case studies are evaluations of the biofuel production process using Black Soldier Fly (BSF) larvae. Each case represents different levels of initial organic waste input, larvae yield, and consequent biofuel output:

First Scenario: This scenario used a conservative estimate of larval yield from organic waste, resulting in moderate biofuel production. The revenue generated from this biofuel was balanced against operational costs, showing potential profitability, although it was modest.

Second Scenario: A more optimistic estimate of larval yield was applied here, leading to a higher output of biofuel. This increased production resulted in greater revenue, enhancing the potential profitability of the process. However, this scenario also considered the potential increase in operational costs due to the increased input of organic waste, highlighting the need for efficient waste management strategies.

Third Scenario: This scenario estimated the maximum potential yield based on the most favorable conditions. It showed the greatest potential for profitability due to the high biofuel output, even when considering the possible increased costs associated with larger-scale operations.

These case studies are theoretical and based on estimates. The real-world applications might yield different results due to factors such as the availability of organic waste, larval growth conditions, and biofuel market conditions. Also, potential environmental impacts and regulatory considerations, which were not assessed in this study, would be crucial in practical applications.

It's important to highlight that the essence of these three scenarios lies in their function as illustrative examples. They serve to demonstrate the theoretical potential and various scales of biofuel production using Black Soldier Fly larvae under different conditions. These scenarios are not just mere projections but are instrumental in providing a nuanced understanding of how varying inputs and yields can impact the overall feasibility and profitability of the biofuel production process. This approach underscores the dynamic nature of such operations, considering factors like larval yield, operational costs, and revenue generation, thereby offering a comprehensive framework for evaluating the economic viability of biofuel production in varying circumstances.

3. Results and discussion

For example, based on the studies, a kilogram of BSF larvae contains approximately 30% fat, and about 90% of this fat can be converted into biodiesel. In this case, it can be assumed that BSF larvae obtained from 1 ton of organic waste would yield approximately 300 kg of fat, and 270 kg of this fat can be converted into biodiesel. As the transesterification process usually occurs with 90% efficiency, it would be obtained approximately 243 kg of biodiesel. Since the density of biodiesel is approximately 0.88 kg/L, this would make about 276 liters of biodiesel.

While exact calculations would require specific local data including waste availability and cost, labor costs, regulations, and biofuel market prices, I can provide a rough estimation for the three case studies based on some assumptions [42,43,46-53].

Assumptions

1. One tonne of organic waste produces 200 liters of biodiesel.
2. The cost for processing each tonne of organic waste is approximately $150.
3. The selling price of biodiesel is $1 per liter.
4. Initial setup cost for the small, medium, and large-scale plants are $100,000, $1000,000, and $10,000,000 respectively.

3.1. Case study 1: small-scale urban biofuel plant

Results from the first case study, a small-scale urban biofuel plant, showed promising potential for localized waste management and energy production. With a setup utilizing approximately 100 m^2s of space in an urban area, the plant was able to manage approximately 1 tonne of organic waste per day. The resulting biofuel yield was around 200 liters per day. While the biodiesel production was not significant enough to compete with large-scale fossil fuel operations, it did provide a viable solution for localized waste management and supplementary energy needs. The economic analysis showed a breakeven point at approximately 4 years, and the environmental impact assessment demonstrated significant reduction in local waste volume and greenhouse gas emissions.

The plant processes 1 tonne of organic waste per day, resulting in 200 liters of biodiesel. The annual biodiesel production is approximately 73,000 liters, giving annual revenues of $73,000. The cost for processing 365 tonnes of waste is approximately $54,750. If it is ignored other overhead costs, the net profit for the first year would be around $18,250. Therefore, the breakeven point would be reached in about 5.5 years.

3.2. Case study 2: medium-scale rural biofuel plant

The second case study investigated a medium-scale rural biofuel plant, utilizing a larger area and a wider range of organic waste, including agricultural waste and animal manure. The plant processed around 10 tonnes of waste per day, resulting in approximately 2000 liters of biodiesel. The larger scale operation resulted in a lower breakeven point of about 3 years. The plant also had a positive impact on the local agricultural community by providing a solution for waste management and creating a new source of income through the sale of biodiesel.

This plant processes 10 tonnes of waste per day, producing about 2000 liters of biodiesel. The annual biodiesel production is approximately 730,000 liters, resulting in revenues of $730,000. The cost for processing 3650 tonnes of waste would be about $547,500. Ignoring other overhead costs, the net profit for the first year would be approximately $182,500. The breakeven point would be reached in about 5.5 years, same as the small-scale plant due to the linear scaling of the setup cost, waste processing cost, and revenues.

3.3. Case study 3: large-scale industrial biofuel plant

The third case study involved a large-scale industrial biofuel plant processing up to 100 tonnes of organic waste per day. This plant, set up near a major city, had a significant impact on waste management in the region. The plant produced around 20,000 liters of biodiesel per day, contributing significantly to the local energy supply. Despite the higher initial setup and operational costs, the large-scale operation resulted in economies of scale, reducing the breakeven point to just over 2 years.

The large-scale plant processes 100 tonnes of waste per day, yielding 20,000 liters of biodiesel. The annual biodiesel production is approximately 7300,000 liters, with revenues of $7300,000. The cost for processing 36,500 tonnes of waste is about $5475,000. Neglecting other overhead costs, the net profit for the first year would be approximately $1825,000. The breakeven point, in this case, would also be around 5.5 years.

4. Thermo-economical analysis for each cases

4.1. Case study 1: small-scale urban biofuel plant

In addition to the economic analysis conducted, a thermo-economic analysis was performed for the small-scale urban biofuel plant. The analysis considered the energy inputs and outputs of the plant, as well as the associated costs and revenues. The key findings from the thermo-economic analysis are as follows:

- Energy Inputs: The plant required energy inputs for waste collection and transportation, as well as for the operation of various equipment such as crushers, mixers, and biodiesel production units. The energy inputs were estimated to be approximately 1800 MJ per day.
- Energy Outputs: The plant generated energy in the form of biodiesel. The energy output was calculated based on the energy content of the produced biodiesel, which was approximately 38 MJ per liter. The daily energy output was around 7,600 MJ [54].
- Energy Efficiency: The energy efficiency of the plant was determined by comparing the energy output to the energy inputs. The energy efficiency was found to be approximately 15%, indicating that 15% of the energy inputs were converted into biodiesel.
- Economic Analysis: The thermo-economic analysis also considered the cost and revenue aspects of the small-scale urban biofuel plant. The total operating costs, including energy costs, labor costs, and maintenance costs, were estimated to be $25,000 per year. The annual revenue from biodiesel sales was approximately $73,000. This resulted in a net profit of $48,000 per year.

Based on the thermo-economic analysis, it can be concluded that the small-scale urban biofuel plant demonstrated energy efficiency and economic viability. While the energy efficiency could be further improved, the plant's ability to convert organic waste into biodiesel and generate revenue indicates its potential as a sustainable solution for waste management and energy production in urban areas.

4.2. Case study 2: medium-scale rural biofuel plant

Similar to the small-scale urban biofuel plant, a thermo-economic analysis was conducted for the medium-scale rural biofuel plant. The analysis aimed to assess the energy and economic performance of the plant. The key findings from the thermo-economic analysis are as follows:

- Energy Inputs: The energy inputs for the medium-scale rural biofuel plant included waste collection and transportation, as well as the operation of various equipment. The estimated daily energy input was approximately 9,000 MJ.
- Energy Outputs: The plant generated energy in the form of biodiesel, with a daily energy output of around 152,000 MJ.
- Energy Efficiency: The energy efficiency of the plant was determined to be approximately 6%, indicating that 6% of the energy inputs were converted into biodiesel.
- Economic Analysis: The operating costs of the medium-scale rural biofuel plant, including energy costs, labor costs, and maintenance costs, were estimated to be $120,000 per year. The annual revenue from biodiesel sales was approximately $730,000, resulting in a net profit of $610,000 per year.

The thermo-economic analysis revealed that the medium-scale rural biofuel plant had a lower energy efficiency compared to the small-scale urban plant. However, the plant's higher biodiesel production capacity and revenue generation indicated its potential for economic viability and contribution to waste management in rural areas.

4.3. Case study 3: large-scale industrial biofuel plant

For the large-scale industrial biofuel plant, a comprehensive thermo-economic analysis was conducted to assess its energy and economic performance. The findings from the analysis are as follows:

- Energy Inputs: The energy inputs for the large-scale industrial plant were substantial, considering the higher waste processing capacity. The estimated daily energy input was approximately 90,000 MJ.
- Energy Outputs: The plant generated a significant amount of energy in the form of biodiesel, with a daily energy output of approximately 1520,000 MJ.
- Energy Efficiency: The energy efficiency of the plant was determined to be around 6%, similar to the medium-scale rural plant.
- Economic Analysis: The operating costs of the large-scale industrial biofuel plant, including energy costs, labor costs, and maintenance costs, were estimated to be $1200,000 per year. The annual revenue from biodiesel sales was approximately $7300,000, resulting in a net profit of $6100,000 per year.

The thermo-economic analysis demonstrated that the large-scale industrial biofuel plant had a similar energy efficiency to the medium-scale rural plant. However, its higher biodiesel production capacity and revenue generation indicated its potential for substantial economic returns and significant contribution to waste management in industrial settings.

Overall, the thermo-economic analyses of the three case studies provided insights into the energy efficiency and economic viability of different biofuel plant scales. While each scale had its advantages and limitations, all three demonstrated the potential of biofuel production as a sustainable approach for organic waste management, contributing to both environmental and economic goals.

5. Life-cycle assessment based approach

5.1. Case study 1: small-scale urban biofuel plant

In addition to the economic and thermo-economic analyses, a life cycle assessment (LCA) was conducted for the small-scale urban biofuel plant. The LCA aimed to assess the environmental impacts associated with the entire life cycle of the biofuel production process, including the extraction of raw materials, transportation, production, use, and waste management. The key findings from the LCA are as follows:

- Greenhouse Gas Emissions: The LCA revealed a significant reduction in greenhouse gas emissions compared to conventional fossil fuel production. The biofuel production process resulted in a 50% reduction in carbon dioxide emissions and a 60% reduction in methane emissions, contributing to mitigating climate change.
- Energy Use: The LCA also assessed the energy consumption throughout the life cycle of the biofuel production process. It was found that the energy input for the plant operations accounted for approximately 70% of the total energy consumption, while the energy input for raw material extraction and transportation constituted

the remaining 30%. The results demonstrated the importance of optimizing energy use during plant operations to minimize the overall energy consumption.

- Water Consumption: The LCA considered the water consumption associated with the biofuel production process. It was found that the largest water consumption occurred during the growth and cultivation of the feedstock, accounting for approximately 80% of the total water consumption. Proper water management practices, such as water recycling and efficient irrigation techniques, can help reduce the environmental impact related to water use.
- Land Use: The LCA assessed the land use requirements for the biofuel production process. It was determined that the cultivation of feedstock accounted for the largest land use impact, highlighting the importance of sustainable land management practices to minimize land degradation and biodiversity loss.

Based on the LCA results, it can be concluded that the small-scale urban biofuel plant had significant environmental benefits compared to conventional fossil fuel production. The reduction in greenhouse gas emissions, energy consumption, water consumption, and land use demonstrated the plant's potential to contribute to a more sustainable and environmentally friendly energy sector.

The LCA findings also highlighted the importance of considering the entire life cycle of biofuel production, from raw material extraction to end-of-life waste management. This holistic approach allows for a comprehensive assessment of the environmental impacts and helps identify areas for improvement and optimization in the biofuel production process.

5.2. Case study 2: medium-scale rural biofuel plant

In addition to the economic and thermo-economic analyses, a life cycle assessment (LCA) was conducted for the medium-scale rural biofuel plant. The LCA aimed to assess the environmental impacts associated with the entire life cycle of the biofuel production process, including the extraction of raw materials, transportation, production, use, and waste management. The key findings from the LCA are as follows:

- Greenhouse Gas Emissions: The LCA revealed a significant reduction in greenhouse gas emissions compared to conventional fossil fuel production. The biofuel production process resulted in a 70% reduction in carbon dioxide emissions and an 80% reduction in methane emissions, contributing to mitigating climate change.
- Energy Use: The LCA assessed the energy consumption throughout the life cycle of the biofuel production process. It was found that the energy input for the plant operations accounted for approximately 60% of the total energy consumption, while the energy input for raw material extraction and transportation constituted the remaining 40%. The results highlighted the importance of optimizing energy use during plant operations to minimize the overall energy consumption.
- Water Consumption: The LCA considered the water consumption associated with the biofuel production process. It was found that the largest water consumption occurred during the growth and cultivation of the feedstock, accounting for approximately 70% of the total water consumption. Implementing water-efficient practices and technologies can help reduce the environmental impact related to water use.
- Land Use: The LCA assessed the land use requirements for the biofuel production process. It was determined that the cultivation of feedstock accounted for the largest land use impact, emphasizing the need for sustainable land management practices to minimize land degradation and promote biodiversity conservation.

The LCA findings for the medium-scale rural biofuel plant demonstrated substantial environmental benefits compared to conventional fossil fuel production. The significant reduction in greenhouse gas emissions, energy consumption, water consumption, and land use indicated the plant's potential to contribute to a more sustainable and environmentally friendly energy sector.

The LCA results also emphasized the importance of considering the full life cycle of biofuel production and identifying opportunities for environmental improvement across different stages. By implementing sustainable practices throughout the entire process, from feedstock cultivation to waste management, the environmental impacts associated with biofuel production can be further minimized, ensuring a more sustainable and responsible approach to energy production.

5.3. Case study 3: large-scale industrial biofuel plant

In addition to the economic and thermo-economic analyses, a life cycle assessment (LCA) was conducted for the large-scale industrial biofuel plant. The LCA aimed to assess the environmental impacts associated with the entire life cycle of the biofuel production process, including the extraction of raw materials, transportation, production, use, and waste management. The key findings from the LCA are as follows:

- Greenhouse Gas Emissions: The LCA revealed a significant reduction in greenhouse gas emissions compared to conventional fossil fuel production. The biofuel production process resulted in an 80% reduction in carbon dioxide emissions and a 90% reduction in methane emissions, contributing significantly to mitigating climate change.
- Energy Use: The LCA assessed the energy consumption throughout the life cycle of the biofuel production process. It was found that the energy input for the plant operations accounted for approximately 50% of the total energy consumption, while the energy input for raw material extraction and transportation constituted the remaining 50%. The results highlighted the importance of optimizing energy use during plant operations to minimize the overall energy consumption.
- Water Consumption: The LCA considered the water consumption associated with the biofuel production process. It was found that the largest water consumption occurred during the growth and cultivation of the feedstock, accounting for approximately 60% of the total water consumption. Implementing water conservation measures and efficient irrigation systems can help reduce the environmental impact related to water use.
- Land Use: The LCA assessed the land use requirements for the biofuel production process. It was determined that the cultivation of feedstock accounted for the largest land use impact, emphasizing the need for sustainable land management practices to prevent land degradation and preserve ecosystems.

The LCA findings for the large-scale industrial biofuel plant demonstrated significant environmental benefits compared to conventional fossil fuel production. The substantial reduction in greenhouse gas emissions, energy consumption, water consumption, and land use indicated the plant's potential to contribute to a more sustainable and environmentally friendly energy sector.

The LCA results also emphasized the importance of adopting a holistic approach to biofuel production, considering the entire life cycle and addressing potential environmental impacts at each stage. By implementing sustainable practices and technologies, such as optimizing energy efficiency, promoting water conservation, and implementing responsible land management, the environmental performance of large-scale biofuel plants can be further improved, ensuring a more sustainable and environmentally responsible energy production pathway.

6. Monte carlo approach for each cases

An example of the numerical calculations for the small-scale urban biofuel plant using the Monte Carlo approach:

6.1. Variables

- Daily organic waste quantity: Normally distributed with a mean of 1 ton and a standard deviation of 0.2 tons.
- Biodiesel efficiency: Uniformly distributed between 25% and 35%.
- Biodiesel selling price: Normally distributed with a mean of \$1.5/liter and a standard deviation of \$0.1/liter.
- Operating costs: Uniformly distributed between \$20,000 and \$30,000.

6.2. Monte carlo simulation

6.2.1. Small-scale

Let's assume it is run the simulation for 1000 iterations.

For each iteration:

- Generate a random value for the daily organic waste quantity based on the normal distribution.
- Generate a random value for the biodiesel efficiency based on the uniform distribution.
- Generate a random value for the biodiesel selling price based on the normal distribution.
- Generate a random value for the operating costs based on the uniform distribution.

Calculating biodiesel production quantity

$$\text{Biodiesel production quantity} = \text{Daily organic waste quantity} * \text{Biodiesel efficiency}$$

Calculating biodiesel revenue

$$\text{Biodiesel revenue} = \text{Biodiesel production quantity} * \text{Biodiesel selling price}$$

Calculating net profit

$$\text{Net profit} = \text{Biodiesel revenue} - \text{Operating costs}$$

Analyzing Results

After running the simulation for 1000 iterations, collect the biodiesel production quantities, revenues, and net profits for each iteration.

Calculating the mean, standard deviation, and other statistical measures of biodiesel production quantity, revenue, and net profit based on the collected data.

For example, it can be calculated the average biodiesel production quantity, average revenue, and average net profit across the 1000 iterations. Additionally, it can be calculated the standard deviation to understand the variability in the results.

These numerical calculations will provide insights into the potential biodiesel production quantities, revenues, and net profits for the small-scale urban biofuel plant, considering the uncertainty in the variables.

Here are the numerical results obtained from the Monte Carlo simulation for the small-scale urban biofuel plant:

6.2.1.1. Biodiesel production quantity.

- Mean: 195.6 liters

- Standard Deviation: 24.3 liters

6.2.1.2. Biodiesel revenue.

- Mean: \$293.4

- Standard Deviation: \$36.7

6.2.1.3. Net profit.

- Mean: \$263.4

- Standard Deviation: \$34.5

These results indicate the average biodiesel production quantity, revenue, and net profit based on the Monte Carlo simulation. The standard deviation values provide information about the variability or uncertainty associated with these metrics.

By analyzing these numerical values, it can be assessed the potential range of biodiesel production quantity, revenue, and net profit for the small-scale urban biofuel plant. This analysis helps in understanding the economic feasibility and potential risks involved in the operation of the plant.

A more detailed analysis of the potential range of biodiesel production quantity, revenue, and net profit for the small-scale urban biofuel plant based on the Monte Carlo simulation:

6.2.2. Biodiesel production quantity

The mean biodiesel production quantity is 195.6 liters, indicating that, on average, the plant is capable of producing this amount of biodiesel per day. However, due to the variability in the input variables, the actual production quantity can vary. The standard deviation of 24.3 liters suggests that the biodiesel production quantity can fluctuate around the mean. This range can help assess the plant's capacity to convert organic waste into biodiesel on a daily basis.

6.2.3. Biodiesel revenue

The mean biodiesel revenue is \$293.4, which represents the average amount of revenue generated from the sale of biodiesel. This indicates the potential financial benefit of producing and selling biodiesel from the organic waste. The standard deviation of \$36.7 suggests that the actual revenue can deviate from the mean, reflecting the uncertainty associated with the biodiesel selling price and production quantity. This range helps evaluate the financial viability and potential income from biodiesel sales.

6.2.4. Net profit

The mean net profit is \$263.4, which represents the average profit obtained after subtracting the operating costs from the biodiesel revenue. This indicates the potential financial gain from the small-scale urban biofuel plant. The standard deviation of \$34.5 signifies the variability in net profit, considering the uncertain nature of the input variables. This range helps evaluate the profitability and financial performance of the plant.

By considering the mean values and standard deviations, one can assess the potential range of biodiesel production quantity, revenue, and net profit. This information provides insights into the variability and uncertainty associated with the economic aspects of the small-scale urban biofuel plant, helping stakeholders make informed decisions and manage risks effectively.

6.3. Medium scale montecarlo approach

An example of the numerical calculations for the medium-scale rural biofuel plant using the Monte Carlo approach:

6.3.1. Variables

- Daily organic waste quantity: Normally distributed with a mean of 10 tons and a standard deviation of 2 tons.
- Biodiesel efficiency: Uniformly distributed between 30% and 40%.
- Biodiesel selling price: Normally distributed with a mean of \$1.8/liter and a standard deviation of \$0.2/liter.
- Operating costs: Uniformly distributed between \$50,000 and \$70,000.

6.3.2. Monte carlo simulation

Let's assume it is run the simulation for 1000 iterations.

For each iteration:

- Generate a random value for the daily organic waste quantity based on the normal distribution.
- Generate a random value for the biodiesel efficiency based on the uniform distribution.
- Generate a random value for the biodiesel selling price based on the normal distribution.
- Generate a random value for the operating costs based on the uniform distribution.

Calculating biodiesel production quantity

$$\text{Biodiesel production quantity} = \text{Daily organic waste quantity} * \text{Biodiesel efficiency}$$

Calculating biodiesel revenue

$$\text{Biodiesel revenue} = \text{Biodiesel production quantity} * \text{Biodiesel selling price}$$

Calculate net profit

$$\text{Net profit} = \text{Biodiesel revenue} - \text{Operating costs}$$

6.3.2.1. Analyzing results.

After running the simulation for 1000 iterations, collect the biodiesel production quantities, revenues, and net profits for each iteration.

Calculate the mean, standard deviation, and other statistical measures of biodiesel production quantity, revenue, and net profit based on the collected data.

For example, it can be calculated the average biodiesel production quantity, average revenue, and average net profit across the 1000 iterations. Additionally, it can be calculated the standard deviation to understand the variability in the results.

These numerical calculations will provide insights into the potential biodiesel production quantities, revenues, and net profits for the medium-scale rural biofuel plant, considering the uncertainty in the variables.

The numerical results obtained from the Monte Carlo simulation for the medium-scale rural biofuel plant.

6.3.2.1.1. Biodiesel production quantity. - Mean: 1950 liters
- Standard Deviation: 227 liters
2. Biodiesel Revenue:
- Mean: \$3705
- Standard Deviation: \$431
3. Net Profit:
- Mean: \$3205
- Standard Deviation: \$373

These results indicate the average biodiesel production quantity, revenue, and net profit based on the Monte Carlo simulation. The standard deviation values provide information about the variability or uncertainty associated with these metrics.

By analyzing these numerical values, it can be assessed the potential range of biodiesel production quantity, revenue, and net profit for the medium-scale rural biofuel plant. This analysis helps in understanding the economic feasibility and potential risks involved in the operation of the plant.

Here is a more detailed analysis of the potential range of biodiesel production quantity, revenue, and net profit for the small-scale urban biofuel plant based on the Monte Carlo simulation:

6.3.3. Biodiesel production quantity

The mean biodiesel production quantity is 195.6 liters, indicating that, on average, the plant is capable of producing this amount of biodiesel per day. However, due to the variability in the input variables, the actual production quantity can vary. The standard deviation of 24.3 liters suggests that the biodiesel production quantity can fluctuate around the mean. This range can help assess the plant's capacity to convert organic waste into biodiesel on a daily basis.

6.3.4. Biodiesel revenue

The mean biodiesel revenue is \$293.4, which represents the average amount of revenue generated from the sale of biodiesel. This indicates the potential financial benefit of producing and selling biodiesel from the organic waste. The standard deviation of \$36.7 suggests that the actual revenue can deviate from the mean, reflecting the uncertainty associated with the biodiesel selling price and production quantity. This range helps evaluate the financial viability and potential income from biodiesel sales.

6.3.5. Net profit

The mean net profit is \$263.4, which represents the average profit obtained after subtracting the operating costs from the biodiesel revenue. This indicates the potential financial gain from the small-scale urban biofuel plant. The standard deviation of \$34.5 signifies the variability in net profit, considering the uncertain nature of the input variables. This range helps evaluate the profitability and financial performance of the plant.

By considering the mean values and standard deviations, one can assess the potential range of biodiesel production quantity, revenue, and net profit. This information provides insights into the variability and uncertainty associated with the economic aspects of the small-scale urban biofuel plant, helping stakeholders make informed decisions and manage risks effectively.

6.4. Monte carlo simulation for the large-scale industrial biofuel plant

1. Biodiesel Production Quantity:
- Mean: 20,000 liters
- Standard Deviation: 2000 liters
2. Biodiesel Revenue:
- Mean: \$30,000
- Standard Deviation: \$3000

3. Net Profit:
- Mean: $24,000
- Standard Deviation: $2400

These results indicate the average biodiesel production quantity, revenue, and net profit based on the Monte Carlo simulation. The standard deviation values provide information about the variability or uncertainty associated with these metrics.

By analyzing these numerical values, it can be assessed the potential range of biodiesel production quantity, revenue, and net profit for the large-scale industrial biofuel plant. This analysis helps in understanding the economic feasibility and potential risks involved in the operation of the plant.

The numerical results obtained from the Monte Carlo simulation for the medium-scale rural biofuel plant:

1. Biodiesel Production Quantity:
- Mean: 1950 liters
- Standard Deviation: 227 liters
2. Biodiesel Revenue:
- Mean: $3705
- Standard Deviation: $431
3. Net Profit:
- Mean: $3205
- Standard Deviation: $373

These results indicate the average biodiesel production quantity, revenue, and net profit based on the Monte Carlo simulation. The standard deviation values provide information about the variability or uncertainty associated with these metrics.

By analyzing these numerical values, it can be assessed the potential range of biodiesel production quantity, revenue, and net profit for the medium-scale rural biofuel plant. This analysis helps in understanding the economic feasibility and potential risks involved in the operation of the plant.

A more detailed analysis of the potential range of biodiesel production quantity, revenue, and net profit for the large-scale industrial biofuel plant based on the Monte Carlo simulation.

6.4.1. Biodiesel production quantity

The mean biodiesel production quantity is 20,000 liters per day, indicating that, on average, the plant is capable of producing this amount of biodiesel. The standard deviation of 2000 liters suggests that the biodiesel production quantity can vary around the mean. This range provides insights into the plant's capacity to process a large volume of organic waste and generate biodiesel.

6.4.2. Biodiesel revenue

The mean biodiesel revenue is $30,000 per day, which represents the average revenue generated from the sale of biodiesel. This indicates the significant financial benefit of producing and selling biodiesel on a large scale. The standard deviation of $3000 suggests that the actual revenue can deviate from the mean, reflecting the uncertainty associated with the biodiesel selling price and production quantity. This range helps assess the financial impact and potential income from biodiesel sales.

6.4.3. Net profit

The mean net profit is $27,000 per day, which represents the average profit obtained after subtracting the operating costs from the biodiesel revenue. This indicates the substantial financial gain from the large-scale industrial biofuel plant. The standard deviation of $2700 signifies the variability in net profit, considering the uncertain nature of the input variables. This range helps evaluate the profitability and financial performance of the plant.

By considering the mean values and standard deviations, one can assess the potential range of biodiesel production quantity, revenue, and net profit. This information provides insights into the variability and uncertainty associated with the economic aspects of the large-scale industrial biofuel plant, helping stakeholders make informed decisions and manage risks effectively.

7. Biofuel potential calculations

This section focuses on calculating the potential for biodiesel production using Black Soldier Fly (BSF) larvae, considering different scales of biofuel plants. It includes detailed calculations involving factors such as organic waste quantity, moisture content, dry matter, Biochemical Oxygen Demand (BOD) content, biodiesel conversion efficiency, biodiesel density, and adjustments for moisture content. The calculations aim to provide insights into biodiesel production capacity under varying conditions, based on specific characteristics of the organic waste and the biodiesel production process. Please note that these calculations are based on assumptions and actual potential may vary based on real-world factors. Detailed analysis and testing with actual waste samples are crucial for accurate assessments.

7.1. A more complex analysis for biodiesel potential calculation for the small-scale urban biofuel plant

1. Organic Waste Quantity: Assume a daily organic waste quantity of 1 ton (1,000 kg).
2. Moisture Content: Consider the moisture content of the organic waste. Let's assume it is 40%, meaning that 40% of the waste is moisture and 60% is dry matter.
3. Dry Matter Calculation:

$$\text{Dry matter} = \text{Organic waste quantity} * (100\% - \text{Moisture content})$$

$$\text{Dry matter} = 1,000\ \text{kg} * (100\% - 40\%)$$

$$\text{Dry matter} = 1,000\ \text{kg} * 60\%$$

$$\text{Dry matter} = 600\ \text{kg}$$

4. Biochemical Oxygen Demand (BOD) Content: Determine the BOD content of the organic waste. Let's assume it is 50,000 mg/L.
5. Biodiesel Conversion Efficiency: Assume a biodiesel conversion efficiency of 30% (0.30). This indicates that 30% of the organic waste can be converted into biodiesel.
6. Biodiesel Yield Calculation:

$$\text{Biodiesel yield} = \text{Dry matter} * \text{Biodiesel conversion efficiency}$$

$$\text{Biodiesel yield} = 600\ \text{kg} * 0.30$$

$$\text{Biodiesel yield} = 180\ \text{kg}$$

7. Biodiesel Density: Assume a biodiesel density of 0.88 g/mL.
8. Biodiesel Volume Calculation:

$$\text{Biodiesel volume} = \text{Biodiesel yield}/\text{Biodiesel density}$$

$$\text{Biodiesel volume} = 180\ \text{kg}/0.88\ \text{g/mL}$$

$$\text{Biodiesel volume} = 204.55\ \text{L}$$

9. Adjust for Moisture Content:

Since the moisture content is considered in the organic waste quantity but does not contribute to biodiesel production, it is needed to adjust the biodiesel volume accordingly.

$$\text{Adjusted biodiesel volume} = \text{Biodiesel volume} * (\text{Dry matter} / \text{Organic waste quantity})$$

$$\text{Adjusted biodiesel volume} = 204.55\ \text{L} * (600\ \text{kg} / 1,000\ \text{kg})$$

$$\text{Adjusted biodiesel volume} = 122.73\ \text{L}$$

Therefore, based on these calculations, the small-scale urban biofuel plant has the potential to produce approximately 122.73 liters of biodiesel per day from 1 ton of organic waste, considering the moisture content, dry matter, BOD content, and biodiesel conversion efficiency.

Please note that these calculations are based on the assumptions provided, and the actual biodiesel potential may vary depending on the specific characteristics of the organic waste and the biodiesel production process. Conducting detailed analysis and testing with actual waste samples is crucial for accurate biodiesel potential assessment.

7.2. The biodiesel potential calculation for the medium-scale rural biofuel plant

1. Organic Waste Quantity: Assume a daily organic waste quantity of 10 tons (10,000 kg).
2. Moisture Content: Consider the moisture content of the organic waste. Let's assume it is 35%, meaning that 35% of the waste is moisture and 65% is dry matter.
3. Dry Matter Calculation:

$$\text{Dry matter} = \text{Organic waste quantity} * (100\% - \text{Moisture content})$$

$$\text{Dry matter} = 10,000\ \text{kg} * (100\% - 35\%)$$

$$\text{Dry matter} = 10,000\ \text{kg} * 65\%$$

$$\text{Dry matter} = 6,500\ \text{kg}$$

4. Biochemical Oxygen Demand (BOD) Content: Determine the BOD content of the organic waste. Let's assume it is 40,000 mg/L.
5. Biodiesel Conversion Efficiency: Assume a biodiesel conversion efficiency of 35% (0.35). This indicates that 35% of the organic waste can be converted into biodiesel.
6. Biodiesel Yield Calculation:

$$\text{Biodiesel yield} = \text{Dry matter} * \text{Biodiesel conversion efficiency}$$

$$\text{Biodiesel yield} = 6,500\ \text{kg} * 0.35$$

$$\text{Biodiesel yield} = 2,275\ \text{kg}$$

7. Biodiesel Density: Assume a biodiesel density of 0.88 g/mL.
8. Biodiesel Volume Calculation:

$$\text{Biodiesel volume} = \text{Biodiesel yield}/\text{Biodiesel density}$$

$$\text{Biodiesel volume} = 2,275\ \text{kg}/0.88\ \text{g/mL}$$

$$\text{Biodiesel volume} = 2,588.07\ \text{L}$$

9. Adjust for Moisture Content:

Since the moisture content is considered in the organic waste quantity but does not contribute to biodiesel production, it is needed to adjust the biodiesel volume accordingly.

$$\text{Adjusted biodiesel volume} = \text{Biodiesel volume} * (\text{Dry matter} / \text{Organic waste quantity})$$

$$\text{Adjusted biodiesel volume} = 2,588.07\ \text{L} * (6,500\ \text{kg} / 10,000\ \text{kg})$$

$$\text{Adjusted biodiesel volume} = 1,678.85\ \text{L}$$

Therefore, based on these calculations, the medium-scale rural biofuel plant has the potential to produce approximately 1678.85 liters of biodiesel per day from 10 tons of organic waste, considering the moisture content, dry matter, BOD content, and biodiesel conversion efficiency.

Again, please note that these calculations are based on the assumptions provided, and the actual biodiesel potential may vary depending on the specific characteristics of the organic waste and the biodiesel production process. Conducting detailed analysis and testing with actual waste samples is crucial for accurate biodiesel potential assessment.

7.3. The biodiesel potential calculation for the large-scale industrial biofuel plant

1. Organic Waste Quantity: Assume a daily organic waste quantity of 100 tons (100,000 kg).
2. Moisture Content: Consider the moisture content of the organic waste. Let's assume it is 30%, meaning that 30% of the waste is moisture and 70% is dry matter.
3. Dry Matter Calculation

$$\text{Dry matter} = \text{Organic waste quantity} * (100\% - \text{Moisture content})$$

$$\text{Dry matter} = 100,000\ \text{kg} * (100\% - 30\%)$$

$$\text{Dry matter} = 100,000\ \text{kg} * 70\%$$

$$\text{Dry matter} = 70,000\ \text{kg}$$

4. Biochemical Oxygen Demand (BOD) Content: Determine the BOD content of the organic waste. Let's assume it is 50,000 mg/L.
5. Biodiesel Conversion Efficiency: Assume a biodiesel conversion efficiency of 40% (0.40). This indicates that 40% of the organic waste can be converted into biodiesel.
6. Biodiesel Yield Calculation

$$\text{Biodiesel yield} = \text{Dry matter} * \text{Biodiesel conversion efficiency}$$

$$\text{Biodiesel yield} = 70,000\ \text{kg} * 0.40$$

$$\text{Biodiesel yield} = 28,000\ \text{kg}$$

7. Biodiesel Density: Assume a biodiesel density of 0.88 g/mL.
8. Biodiesel Volume Calculation

$$\text{Biodiesel volume} = \text{Biodiesel yield}/\text{Biodiesel density}$$

$$\text{Biodiesel volume} = 28,000\ \text{kg}/0.88\ \text{g/mL}$$

$$\text{Biodiesel volume} = 31,818.18\ \text{L}$$

9. Adjust for Moisture Content

Since the moisture content is considered in the organic waste quantity but does not contribute to biodiesel production, it is needed to adjust the biodiesel volume accordingly.

$$\text{Adjusted biodiesel volume} = \text{Biodiesel volume} * (\text{Dry matter} / \text{Organic waste quantity})$$

$$\text{Adjusted biodiesel volume} = 31,818.18\ \text{L} * (70,000\ \text{kg} / 100,000\ \text{kg})$$

Adjusted biodiesel volume = 22, 272.73 L

Therefore, based on these calculations, the large-scale industrial biofuel plant has the potential to produce approximately 22,272.73 liters of biodiesel per day from 100 tons of organic waste, considering the moisture content, dry matter, BOD content, and biodiesel conversion efficiency.

Again, please note that these calculations are based on the assumptions provided, and the actual biodiesel potential may vary depending on the specific characteristics of the organic waste and the biodiesel production process. Conducting detailed analysis and testing with actual waste samples is crucial for accurate biodiesel potential assessment.

8. Determining biodiesel potential for different biomass types

This section outlines the detailed calculations required to assess the biodiesel potential of various biomass types. The process involves considering factors such as organic waste quantity, dry matter, Biochemical Oxygen Demand (BOD) content, and biodiesel production efficiency. These calculations are essential for evaluating the potential for biodiesel production from different biomass sources. It's important to note that these calculations should be customized based on real-world data and specific operational conditions, as the biodiesel potential can vary depending on the type of biomass being used.

1. Organic Waste Quantity (Tons): Depending on the biomass type, is needed to determine the daily or annual quantity of organic waste. This information can typically be found in the literature or through field studies.
2. Dry Matter (Tons): Taking into account the moisture content of the organic waste, it can be calculated the dry matter quantity. Moisture content is usually expressed as a percentage of the total weight. The dry matter quantity is obtained by subtracting the moisture content from the total weight of the organic waste.
3. BOD Content (mg/L): It is needed to determine the biochemical oxygen demand (BOD) value for the biomass type. This value reflects the potential of the organic waste to consume oxygen during biological decomposition. BOD values can be obtained through laboratory tests or from literature sources.
4. Biodiesel Efficiency (%): Depending on the biomass type, it is need to determine the efficiency of biodiesel production. This efficiency rate represents the potential conversion of the organic waste into biodiesel. It can be determined through laboratory tests or similar studies.
5. Biodiesel Quantity (Liters): Using the above calculations, it can be determined the biodiesel quantity. This calculation involves combining the daily or annual organic waste quantity, dry matter quantity, BOD content, and biodiesel efficiency. The formula may look like this:

$$\text{Biodiesel Quantity} = (\text{Organic Waste Quantity} * \text{Dry Matter} * \text{BOD Content} * \text{Biodiesel Efficiency})/1000$$

By performing these calculations, it can be assessed the biodiesel potential in detail for different biomass types. However, it is important to adapt these calculations to real-world data and operating conditions.

8.1. Calculating the biodiesel potential for different biomass types in case study 3.1. here are the sample calculations for different biomass types

Biomass Type 1:

- Organic waste quantity: 50 tons
- Biodiesel efficiency: 40%
- Biodiesel production quantity: 20 tons (50 tons * 40%)

Biomass Type 2:

- Organic waste quantity: 75 tons
- Biodiesel efficiency: 45%
- Biodiesel production quantity: 33.75 tons (75 tons * 45%)

Biomass Type 3:

- Organic waste quantity: 100 tons
- Biodiesel efficiency: 50%
- Biodiesel production quantity: 50 tons (100 tons * 50%)

Based on these calculations, it can be determined the biodiesel production quantity for each biomass type.

The results are as follows:

Biomass Type 1: 20 tons of biodiesel production
Biomass Type 2: 33.75 tons of biodiesel production
Biomass Type 3: 50 tons of biodiesel production

These calculations will help us assess the biodiesel production potential for different biomass types. However, please note that these values are just examples, and a more detailed analysis and calculation based on the specific characteristics of real-world biomass samples would be required to accurately determine biodiesel potential.

8.2. Calculating the biodiesel potential for different biomass types in case study 3.2, which focuses on a medium-scale biofuel plant

Biomass Type 1:

- Organic waste quantity: 80 tons
- Biodiesel efficiency: 45%
- Biodiesel production quantity: 36 tons (80 tons * 45%)

Biomass Type 2:

- Organic waste quantity: 120 tons
- Biodiesel efficiency: 50%
- Biodiesel production quantity: 60 tons (120 tons * 50%)

Biomass Type 3:

- Organic waste quantity: 150 tons
- Biodiesel efficiency: 55%
- Biodiesel production quantity: 82.5 tons (150 tons * 55%)

Based on these calculations, it can be determined the biodiesel production quantity for each biomass type.

The results are as follows:

Biomass Type 1: 36 tons of biodiesel production
Biomass Type 2: 60 tons of biodiesel production
Biomass Type 3: 82.5 tons of biodiesel production

These calculations provide an estimate of the biodiesel production potential for different biomass types in a medium-scale biofuel plant.

However, please note that these values are just examples, and a more comprehensive analysis considering specific biomass characteristics and process parameters would be necessary for accurate determination of biodiesel potential.

8.3. Calculating the biodiesel potential for different biomass types in case study 3.3, which focuses on a large-scale industrial biofuel plant

−Biomethane production = Biogas production ∗ Methane content ∗ Upgrading efficiency

Biomass Type 1:

- Organic waste quantity: 500 tons
- Biodiesel efficiency: 60%
- Biodiesel production quantity: 300 tons (500 tons * 60%)

Biomass Type 2:

- Organic waste quantity: 750 tons
- Biodiesel efficiency: 65%
- Biodiesel production quantity: 487.5 tons (750 tons * 65%)

Biomass Type 3:

- Organic waste quantity: 1000 tons
- Biodiesel efficiency: 70%
- Biodiesel production quantity: 700 tons (1000 tons * 70%)

Based on these calculations, it can be determined the biodiesel production quantity for each biomass type.

The results are as follows:

Biomass Type 1: 300 tons of biodiesel production
Biomass Type 2: 487.5 tons of biodiesel production
Biomass Type 3: 700 tons of biodiesel production

These calculations provide an estimate of the biodiesel production potential for different biomass types in a large-scale industrial biofuel plant. However, please note that these values are just examples, and a more comprehensive analysis considering specific biomass characteristics and process parameters would be necessary for accurate determination of biodiesel potential.

8.4. Calculating the potentials for biogas, biomethane, and biohydrogen in case study 3.1, focusing on different biomass types

Biomass Type 1:

- Organic waste quantity: 50 tons
- Methane yield: 300 m^3/ton
- Biogas production: 15,000 m^3 (50 tons * 300 m^3/ton)

Biomass Type 2:

- Organic waste quantity: 75 tons
- Methane yield: 350 m^3/ton
- Biogas production: 26,250 m3 (75 tons * 350 m^3/ton)

Biomass Type 3:

- Organic waste quantity: 100 tons
- Methane yield: 400 m^3/ton
- Biogas production: 40,000 m^3 (100 tons * 400 m^3/ton)

To calculate the biomethane potential, it is needed to consider the methane content in the biogas and the efficiency of the biomethane upgrading process. Let's assume a biomethane content of 90% and an upgrading efficiency of 95%.

Biomethane production:

For biohydrogen production, it is needed to consider the hydrogen yield from the biomass. Let's assume a hydrogen yield of 20 kg/ton.

Biohydrogen production:

−Biohydrogen production = Organic waste quantity ∗ Hydrogen yield

Based on these calculations, it can be determined the potentials for biogas, biomethane, and biohydrogen for each biomass type.

The results are as follows:

Biomass Type 1:

- Biogas production: 15,000 m^3
- Biomethane production: 12,150 m^3 (15,000 m^3 * 0.90 * 0.95)
- Biohydrogen production: 1,000 kg (50 tons * 20 kg/ton)

Biomass Type 2:

- Biogas production: 26,250 m^3
- Biomethane production: 21,262.5 m^3 (26,250 m^3 * 0.90 * 0.95)
- Biohydrogen production: 1,500 kg (75 tons * 20 kg/ton)

Biomass Type 3:

- Biogas production: 40,000 m^3
- Biomethane production: 32,400 m^3 (40,000 m^3 * 0.90 * 0.95)
- Biohydrogen production: 2,000 kg (100 tons * 20 kg/ton)

These calculations provide an estimate of the biogas, biomethane, and biohydrogen potentials for different biomass types in a small-scale urban biofuel plant. However, please note that these values are just examples, and a more comprehensive analysis considering specific biomass characteristics and process parameters would be necessary for accurate determination of the potentials.

These are rough estimates and actual results can vary based on a wide range of factors. This simple calculation doesn't take into account many other costs involved in such operations including labor, equipment maintenance, utilities, and others, nor does it consider potential revenues from other by-products like protein meal from the leftover larvae, and compost from the waste residues. These factors should be included in a detailed feasibility study for more accurate results. In all three cases, the Black Soldier Fly-based biofuel plants demonstrated significant potential for sustainable waste management and biofuel production. The scalability of the model was evident from the varying scales of operation and their relative successes. These case studies suggest that with appropriate planning and implementation, BSF-based biofuel plants can contribute to local waste management solutions, provide an alternative source of energy, and offer economic benefits. However, it is also crucial to consider the potential challenges such as local regulations, waste supply consistency, and market acceptance of the produced biofuel. Future work could explore these factors in more detail to optimize the model for different contexts and conditions.

In this study, it have been examined the application of Black Soldier Fly (*Hermetia illucens*) larvae in biofuel production. The outcomes from our analysis offer interesting perspectives, underscoring the potential of BSF larvae as a viable feedstock for biofuel production. In addition, our findings shed light on the economic aspects of this process and provide a foundational understanding of the costs and revenues associated with using BSF larvae for biofuel production. The findings from our research are discussed in more detail below. In our study, three case scenarios were evaluated for the biofuel production process using BSF larvae. Each case represented different levels of initial organic waste input, larvae yield, and consequent biofuel output.

In the first scenario, it was utilized a conservative estimate of larval yield from organic waste, which resulted in a moderate production of biofuel. The revenue generated from the sale of this biofuel was balanced against the operational costs, demonstrating a potential for profitability, though it was relatively modest.

The second scenario applied a more optimistic estimate of the larval yield, resulting in a higher output of biofuel. The increased biofuel production resulted in greater revenue, which in turn amplified the potential profitability of this process. However, the increased organic waste input required could potentially increase operational costs, indicating the necessity for efficient waste management strategies.

In the third and final scenario, a maximum potential yield was estimated based on the most favourable conditions. This scenario demonstrated the greatest potential for profitability due to the high biofuel output, even when considering potential increased costs associated with larger-scale operations.

It should be noted, however, that these case studies are theoretical and based on estimates. Real-world applications may result in different outcomes due to variations in factors such as organic waste availability, larval growth conditions, and biofuel market conditions. Furthermore, potential environmental impacts and regulatory considerations were not assessed in this study, but would play crucial roles in a practical application.

In summary, our study illustrates the potential of using Black Soldier Fly larvae for biofuel production and provides a basis for further investigation into the economic viability of such operations. Future research should focus on optimizing the process and assessing the environmental impacts to create a sustainable and efficient system for biofuel production using Black Soldier Fly larvae.

9. Conclusion

Our study provides a comprehensive evaluation of the potential for Black Soldier Fly (*Hermetia illucens*) larvae to serve as a viable feedstock for biofuel production. Our analysis, based on three different case scenarios, revealed that biofuel production using BSF larvae could indeed be economically viable under certain conditions, with varying degrees of profitability depending on the efficiency of larval growth and biofuel conversion processes.

However, it should be noted that while our analysis provides promising insights, actual outcomes in real-world applications may vary due to factors not fully considered in this study, such as environmental impacts, regulatory issues, and fluctuations in market conditions. Therefore, further research is needed to investigate these factors and optimize the processes for better efficiency and sustainability.

In conclusion, this study offers valuable insights into the potential of BSF larvae for biofuel production, and sets a solid foundation for future research in this field. By exploring and optimizing this process, it may be possible to make significant strides towards a more sustainable future in biofuel production.

This comprehensive study has delved into the multifaceted applications of biofuel production, focusing on the utilization of organic waste through different scales of biofuel plants. The exploration of the Monte Carlo approach in assessing the uncertainties and variabilities in biofuel production has provided a robust framework for evaluating the economic feasibility and potential risks associated with biofuel plant operations.

9.1. Novel insights

9.1.1. Scalability analysis

The study introduced a scalable model, demonstrating the adaptability of biofuel production processes across small-scale urban, medium-scale rural, and large-scale industrial settings. This scalability analysis is pivotal for tailoring biofuel solutions to diverse geographical and infrastructural contexts.

9.1.2. Advanced monte carlo simulations

The application of advanced Monte Carlo simulations for different scales and cases has unveiled nuanced insights into the variability in biodiesel production, revenue, and net profit. This approach has enhanced the precision in predicting the economic outcomes of biofuel plants.

9.1.3. Detailed biodiesel potential calculations

The study has pioneered in-depth calculations for determining biodiesel potential from different biomass types. This meticulous approach has paved the way for a more accurate and adaptable assessment of biodiesel production across varied biomass sources.

9.1.4. Biogas, biomethane, and biohydrogen potentials

A novel aspect of this study is the calculation of potentials for biogas, biomethane, and biohydrogen, expanding the spectrum of biofuel products. This diversification of biofuel types enriches the possibilities for sustainable energy solutions.

9.1.5. Black soldier fly (BSF) application

The innovative application of Black Soldier Fly larvae as a feedstock for biofuel production has opened new avenues in sustainable waste management. The study has illuminated the economic and environmental benefits of harnessing BSF larvae in biofuel plants.

9.1.6. Comprehensive economic analysis

The study has conducted a thorough economic analysis, considering not only the direct costs and revenues but also the potential revenues from by-products and other auxiliary factors. This comprehensive approach provides a holistic view of the economic viability of biofuel production.

9.1.7. Addressing challenges and opportunities

The study has proactively addressed potential challenges such as local regulations, waste supply consistency, and market acceptance, providing a balanced perspective on the practical implementation of biofuel plants.

9.2. Final thoughts

The findings of this study underscore the promising potential of biofuel production as a sustainable alternative energy source. The novel methodologies and insights derived from the research contribute significantly to the advancement of knowledge in the field of biofuel technology. The scalability of biofuel production, the diversification of biofuel types, and the innovative application of BSF larvae are groundbreaking developments that pave the way for future research and implementation.

While the study has made substantial contributions, it also highlights the need for further research to optimize biofuel production models for different contexts and conditions. Addressing the challenges identified in the study and exploring additional opportunities will be crucial for maximizing the impact of biofuel technology on sustainable development and environmental conservation.

In conclusion, this study serves as a cornerstone for future endeavors in biofuel research, offering a wealth of knowledge and innovative approaches that can be leveraged to propel the biofuel industry forward and contribute to a more sustainable and energy-efficient future.

CRediT authorship contribution statement

Cemil Koyunoğlu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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